Mini Project 2: 5G-NR channel Interleaver

Date of submission: 10th April 2026

Note

- You may discuss implementation ideas with classmates, but the final submission must reflect your own understanding.
- Identical or nearly identical submissions will get zero marks.
- Write clean, readable code with proper comments rather than overly complex solutions.
- You may be asked to present or explain your code individually. During the presentation, you should be able to i) explain the logic and flow of your program, and ii) justify key design choices.
- To consider your implementation valid, the latency must be less than 11μs, meeting 5G timing requirements, and the resource utilization should not exceed 8% of the total available resources on the Boolean board (part number: xc7s50-csga324-1)

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1 5G-NR Interleaving of coded bits for PUCCH [clause 5.4.1.3 of TS38.212]

Interleaving of coded bits, also known as channel interleaving. The input to the channel interleaving is a sequence of E number of rate matched bits denoted by the vector $\boldsymbol{e} = [e_0, \cdots, e_{E-1}]$. The objective is to rearrange this sequence using a structured write–read procedure to generate an output sequence, $\boldsymbol{f} = [f_0, \cdots, f_{E-1}]$. As shown in Fig. 1, the rearrangement is performed using an isosceles triangular structure with a side length T as follows,

- Determine the smallest integer T such that, $1 + 2 + ... + T = \frac{T(T+1)}{2} \geq E$.
- The isosceles triangular structure will have $\frac{T(T+1)}{2} - E$ number of null bits, as shown in Fig. 1. These null bits are ignored during the writing and reading process, i.e., no bits will be written into those positions.
- The input bit stream $\boldsymbol{e}$ is written into isosceles triangular structure, row-wise. The bits are filled horizontally, starting with the first row and continuing with the subsequent rows.
- The output sequence $\boldsymbol{f}$ is read from the isosceles triangular structure, column-wise. The bits are read vertically, starting with the first and continuing to the subsequent columns.
- The 3GPP procedure for the channel interleaving is given in Algorithm 1 below.

The length of the input bit sequence E is given by,

$$E = 16 \times k, \quad \text{where, } \; k = 1, 2, 3, \ldots, 32$$

So, E takes values as $E \in \{16, 32, 48, 64, \ldots 480, 496, 512\}$

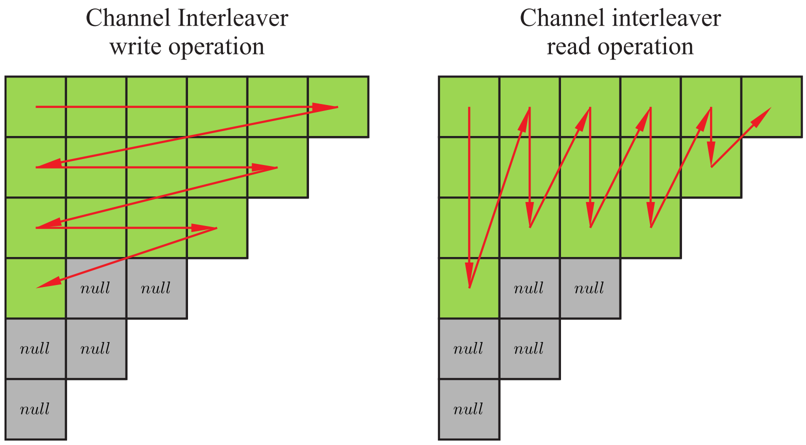

Figure 1: Isosceles triangular structure for the channel interleaver with rate matched output length $E = 16$.

Algorithm 1: 3GPP procedure for interleaving of coded bits

```
// Writing
k = 0;
for i = 0 to T − 1 do
    for j = 0 to T − 1 − i do
        if k < E then
            v_{j,i} = e_k
        ;
        else
            v_{j,i} = NULL
        end
        ;
        k = k + 1;
    end
end
// Reading
k = 0;
for j = 0 to T − 1 do
    for i = 0 to T − 1 − j do
        if v_{j,i} ≠ NULL then
            f_k = v_{j,i};
            k = k + 1;
        end
    end
end
```

1.1 Example

For $E = 16$, the smallest T satisfying $\frac{T(T+1)}{2} \geq E$ is $T = 6$. Total number of positions in the isosceles triangular $\frac{T(T+1)}{2} = 21$. Number of null positions, $21 - 16 = 5$. These null elements are ignored during the row-wise write operation and column-wise read operation, as shown in Fig. 1

1.2 Problem statement

Realize the above channel Interleaving procedure into an IP with the name `channel_interleaver`, using Vivado HLS, with input and output ports being AXI-Stream interfaces. The IP should have

- Input port `inData` with data of width 128 bits and a last signal.
- Output port `outData` with data of width 128 bits and a last signal.
- Control port `cnData` with data of width 64 bits which carries configuration information E which is present in range $[11, 0]$.
- There are 16 test cases that must be successfully passed. The corresponding values of E are provided in the test vector file names. The test vector files required for verification are uploaded under **Mini project 2** in LMS.
- Optimize your implementation as much as you can to reduce latency and resource utilization

1.3 Submission Requirements

1. Prepare a PowerPoint presentation (not more than 7 slides) that includes the following:
    - Answer why channel interleaving is important for 5G-NR PUCCH (not more than 1 slide).
    - **Implementation Overview**: Clearly explain your design approach and implementation methodology.
    - **C/C++ Simulation (CSIM)**: Include a screenshot showing that all test cases pass successfully in a single run.
    - **Synthesis Results**: Include the synthesis report (resource utilization, latency, etc.).
    - **RTL/Cosimulation**: Include a screenshot showing that RTL/Cosim has passed successfully.
    - **Waveform**: Include a waveform screenshot of your implementation.
2. **Files to Submit**: Submit a ZIP file containing, PPT, header file (.h), source code (.cpp), and testbench (.cpp). Name of the ZIP file should be in the format `GroupNumber_RollNumbers.zip`. Any one member of the group can submit the files. Make sure we should be able to replicate the results given in the PPT using the above files.